

CE 3101

Design Of Reinforced Concrete Structures

Design Philosophy

Working Stress Method

- The earliest codified design philosophy is the working stress method of design, which evolved around 1900.
- The traditional method of design not only for RC but also for structural steel and timber design.
- It basically assumes that the structural material behaves in a linear elastic manner and that adequate safety can be ensured by suitably restricting the stresses in the material due to the expected working loads (service loads) on the structure.
- It assumes that both the steel reinforcement and concrete act together and are perfectly elastic at all stages
- The stress-strain relationship of steel and concrete is a straight line under working loads.

Working Stress Method

- The stresses caused by the 'characteristic' or service loads are checked against the permissible (allowable) stress, which is a fraction of the ultimate or yield stress.
- The permissible stress may be defined in terms of a factor of safety, which takes care of the overload or other unknown factors. Thus, the permissible stress is defined by
$$\text{Permissible (allowable) stress} = \text{Ultimate or yield stress} / \text{Factor of safety}$$
- Thus, in WSM, $\text{Working stress} \leq \text{Permissible stress}$

Modular Ratio 'm'

- In the elastic theory, structures having different materials are made equivalent to one common material. In the reinforced concrete structure, concrete and reinforcing steel are, therefore, converted into one material.
- This is done by transformation using the modular ratio m which is the ratio of modulus of elasticity of steel and concrete.
- Thus, $m = E_s/E_c$. where E_s is the modulus of elasticity of steel which is 200000 N/mm^2 . However, concrete has different moduli, as it is not a perfectly elastic material. E_s remains constant, however, E_c changes due to plasticity in concrete.
- The short-term modulus of concrete $E_c = 5000\sqrt{f_{ck}}$ in N/mm^2 , where f_{ck} is the characteristic strength of concrete.
- Accordingly, the modular ratio m is not computed as $m = E_s/E_c = 200000/(5000\sqrt{f_{ck}})$.
- The value of m , as given $280/3\sigma_{cbc}$, partially takes into account long-term effects.

Working Stress Method

TABLE Permissible stresses in concrete (MPa)

Grade of Concrete	Permissible Stress in Compression		Modular Ratio $m = 280/$ $(3\sigma_{cbc})$	Permissible Average Stress in Bond for Plain Bars in Tension* τ_{bd}
	Bending Compression σ_{cbc}	Direct Compression σ_{cc}		
M20	7.0	5.0	13.33	0.8
M25	8.5	6.0	10.98	0.9
M30	10.0	8.0	9.33	1.0
M35	11.5	9.0	8.11	1.1
M40	13.0	10.0	7.18	1.2
M45	14.5	11.0	6.44	1.3
M50	16.0	12.0	5.83	1.4
M55	18.0	13.5	5.19	1.5
M60	20.0	15.0	4.67	1.6

* Bond stress may be increased by 25 per cent for bars in compression.

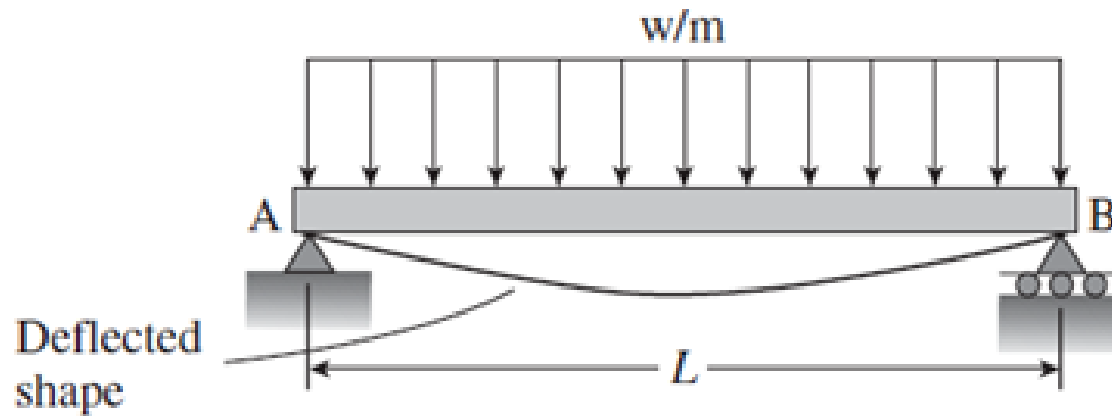
Working Stress Method

TABLE Permissible stresses in steel reinforcement

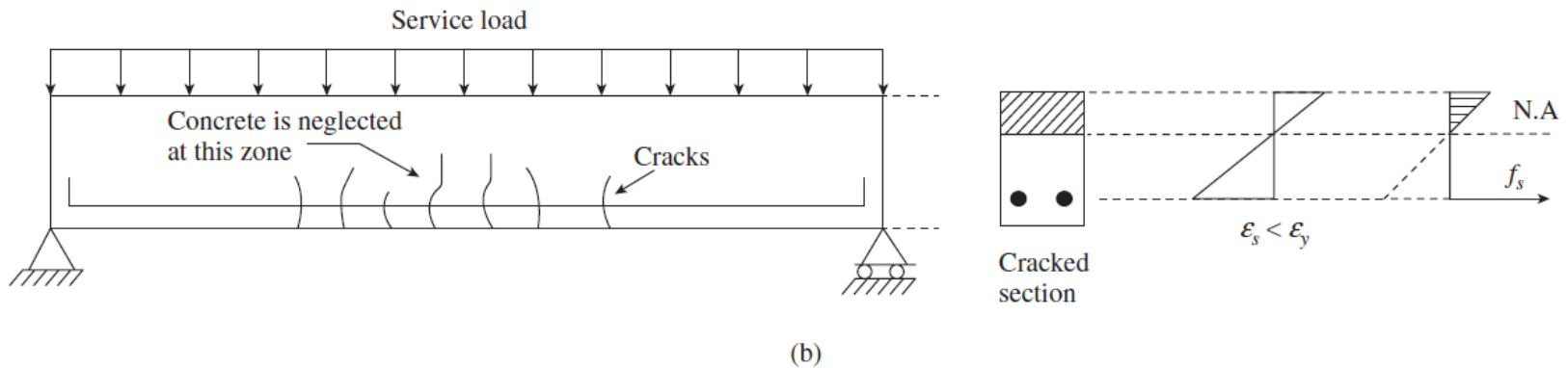
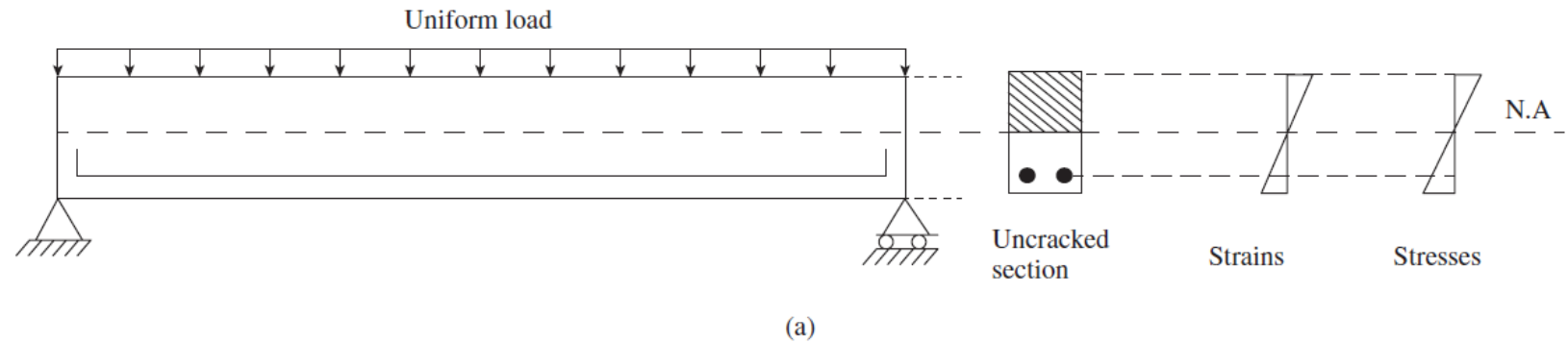
Type of Stress in Steel Reinforcement	Permissible Stress in MPa	
	Mild Steel Bars as per IS 432(Part 1) (Grade Fe 250)	High-yield Strength-deformed Bars as per IS 1786 (Grade Fe 415)
Tension (σ_{st} or σ_{sv})		
Up to and including 20 mm	140	230
Over 20 mm	130	230
Compression in column bars (σ_{sc})	130	190
Compression in beam or slab bars		
Up to and including 20 mm	140	190
Over 20 mm	130	190

Note: For high-yield strength-deformed bars of grade Fe 500, permissible stress in direct tension and flexural tension shall be $0.55f_y$. The permissible stresses for shear and compression steel shall be as for grade Fe 415.

RC Beam under Flexure



Behaviour of an RC beam under different stages of loading



Behaviour of an RC beam under different stages of loading

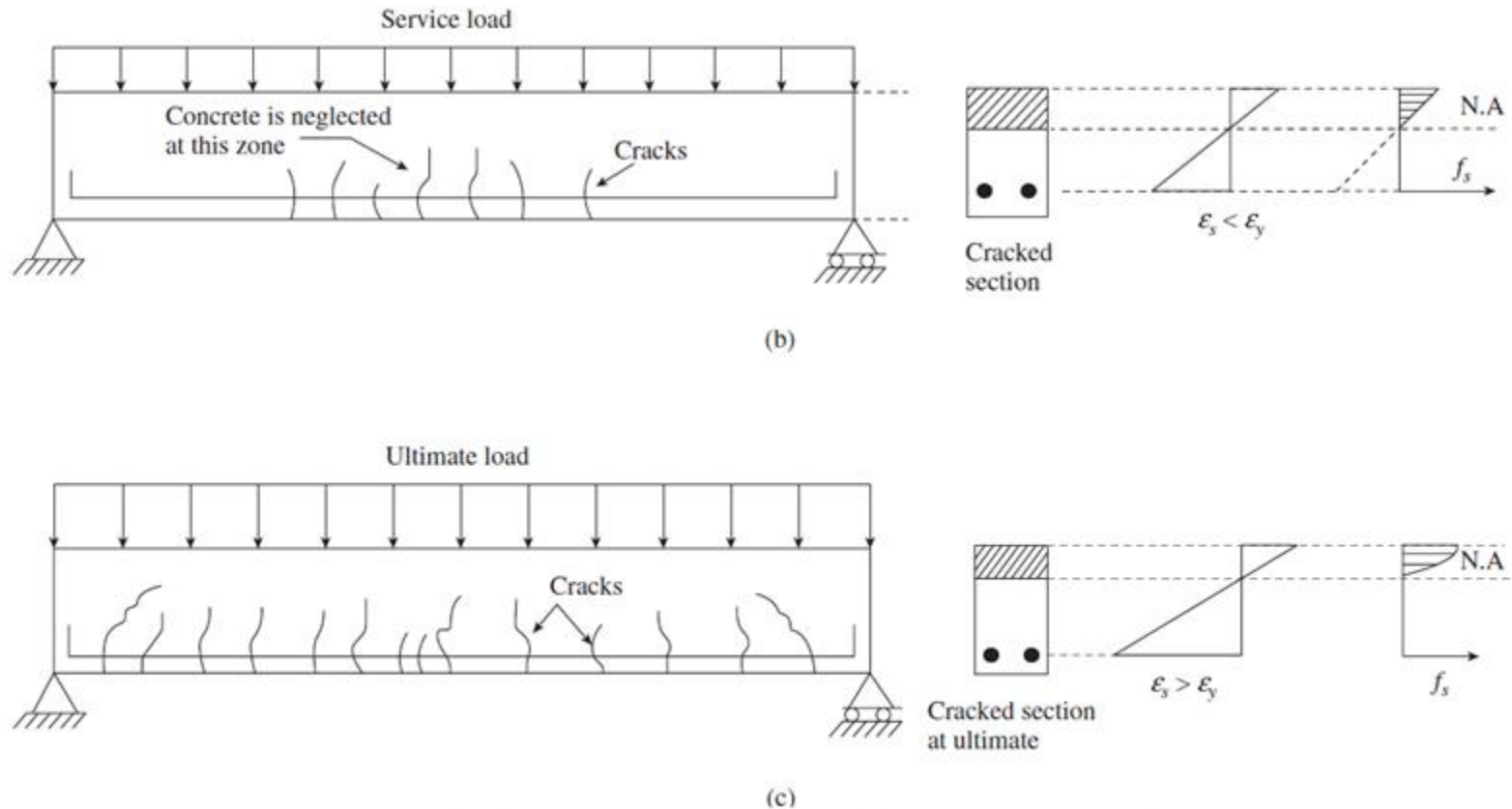


FIG. Behaviour of an RC beam under different stages of loading (a) Before cracking (b) After cracking but before yielding of steel (working load) (c) Ultimate and final stage

Working Stress Method

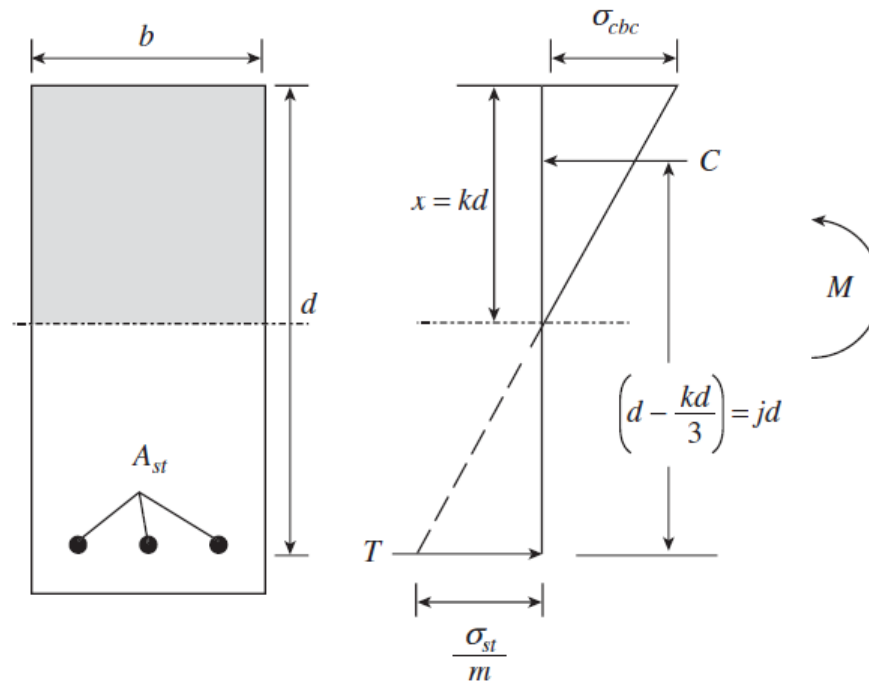


FIG. Stress distribution in rectangular section

Working Stress Method

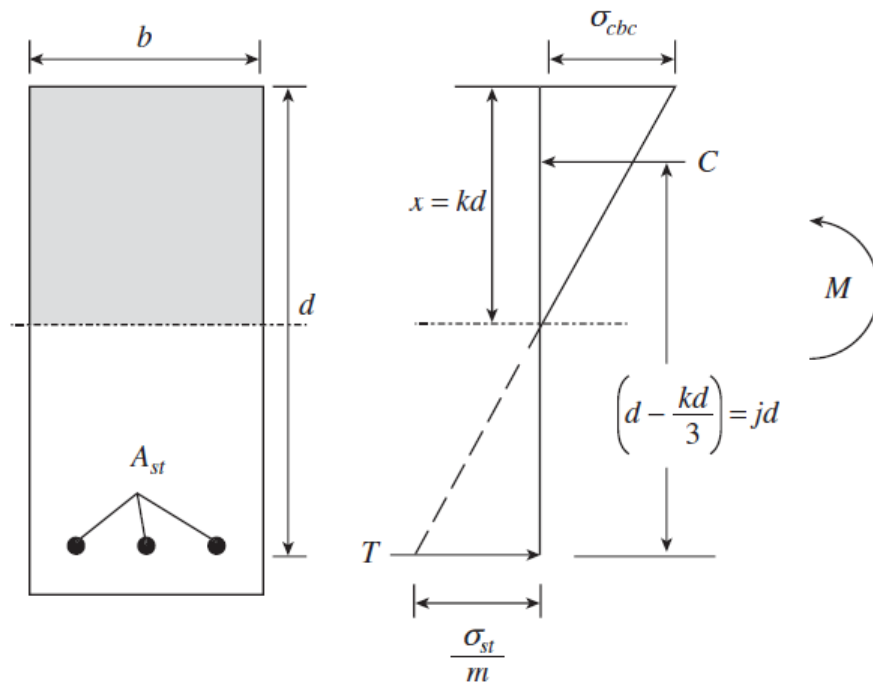


FIG. Stress distribution in rectangular section

C = the total compressive force is given as follows:

$$C = 0.5 \sigma_{cbc} b(kd)$$

b, d = Breadth and effective depth of beam respectively

k = Neutral axis factor

Hence, the lever arm distance is

$$jd = d - \frac{kd}{3} = d \left(1 - \frac{k}{3}\right); \text{ Hence, } j = \left(1 - \frac{k}{3}\right)$$

If the moment at service load is M_s , then

$$M_s = Cjd = 0.5 \sigma_{cbc} b(kd)jd$$

Similarly, taking moments about C yields

$$M_s = Tjd = \sigma_{st} A_{st} jd$$

Using this equation, we get

$$A_{st} = \frac{M_s}{\sigma_{st} jd}$$

Behaviour of an RC beam under different stages of loading

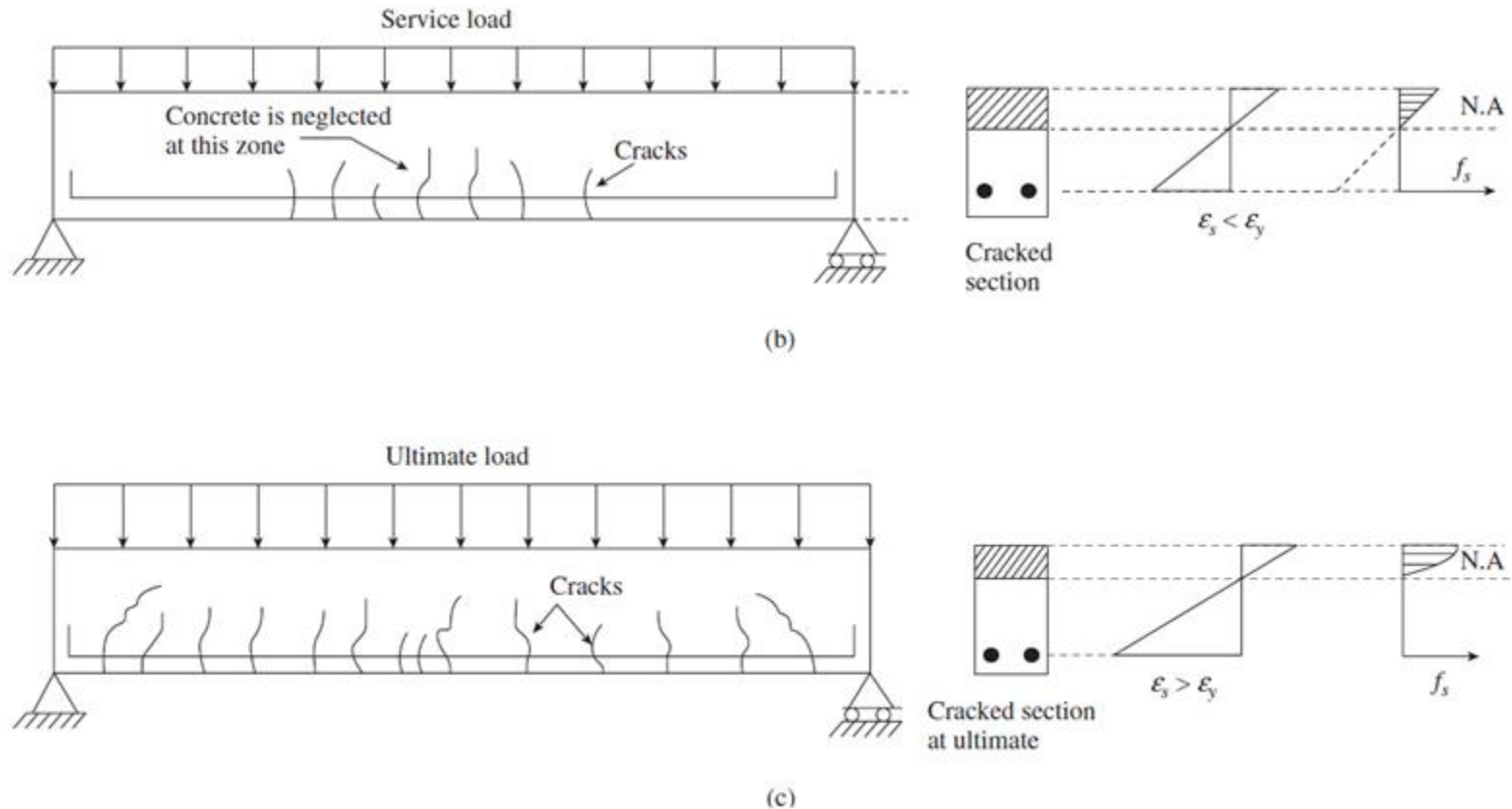


FIG. Behaviour of an RC beam under different stages of loading (a) Before cracking (b) After cracking but before yielding of steel (working load) (c) Ultimate and final stage

Limitations of WSM

- The stress–strain curve for concrete is non-linear and is time dependent. This is particularly so at higher ranges of stress.
- The use of the imaginary concept of modular ratio results in larger percentage of compression steel and generally larger member sizes than the members designed using ultimate load or limit states design.
- WSM does not discriminate between the different types of loads that act simultaneously, but have different degrees of uncertainty.
- In spite of these shortcomings, most structures designed in accordance with this method have performed satisfactorily for many years. Its popularity is due to its simplicity.

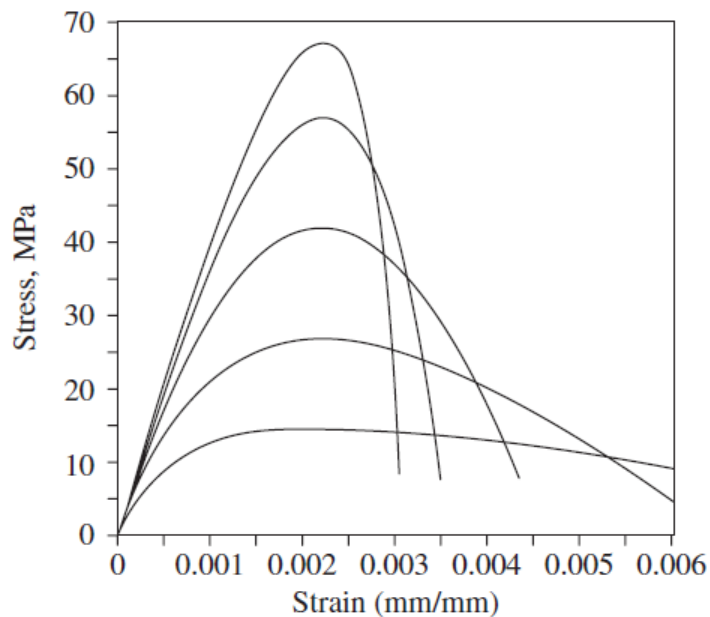
Ultimate Load Design

- The shortcomings of WSM led to the development of the ultimate load design.
- The WSM was followed by the ultimate strength design, which was developed in the 1950s.
- In this method of design, different types of loads are assigned with different load factors under combined loading conditions, thereby overcoming the related shortcomings of WSM.
- A low load factor is used for a load that is known more exactly (e.g., the dead load) and a higher load factor for less-certain loads, (e.g., the live loads).
- In the ultimate load design, the strength of the member must be more than the ultimate load acting on the member.

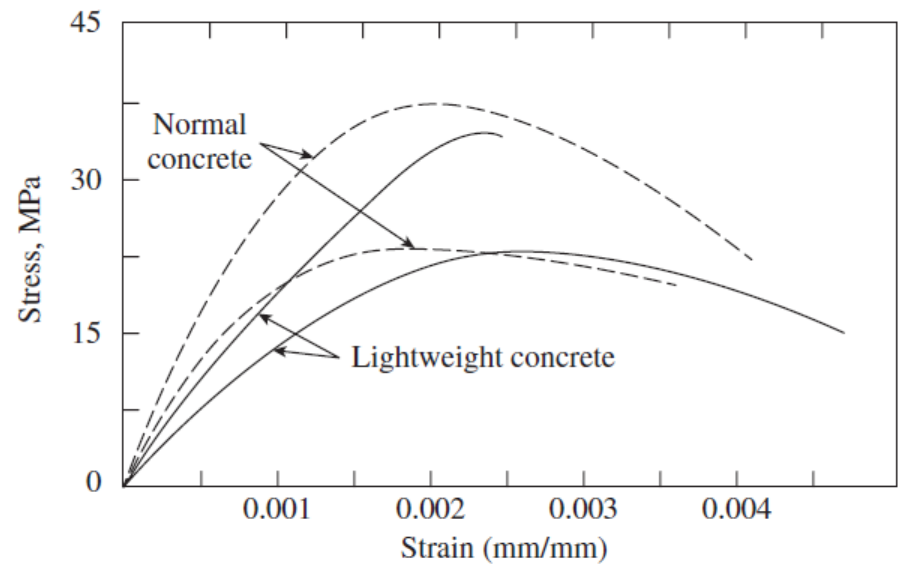
$$\text{Design resistance } (R_n) \geq \text{Design load effect } (\sum \gamma_i Q_i)$$

- One of the disadvantages of this method is that the performance at the normal service loads is not considered.

Stress-Strain Characteristics



(a)



(b)

FIG. Typical stress-strain curves of concrete in compression (a) Concrete with normal weight aggregates (b) Normal weight vs lightweight aggregate concrete

Stress–Strain Characteristics

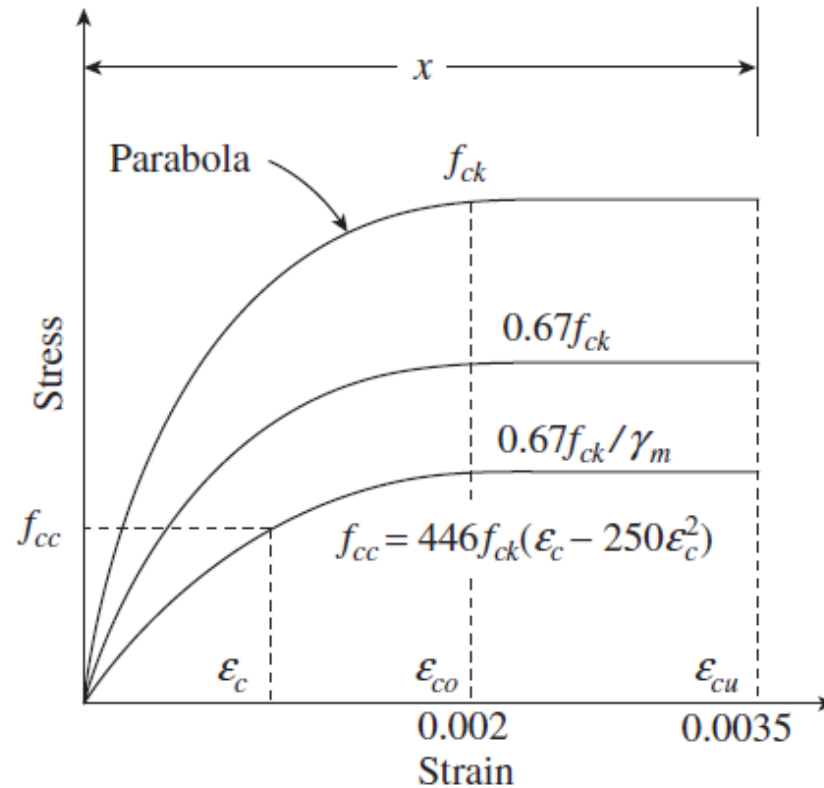


FIG. Idealized stress–Strain curve for concrete

Mechanics of RC beam under flexure

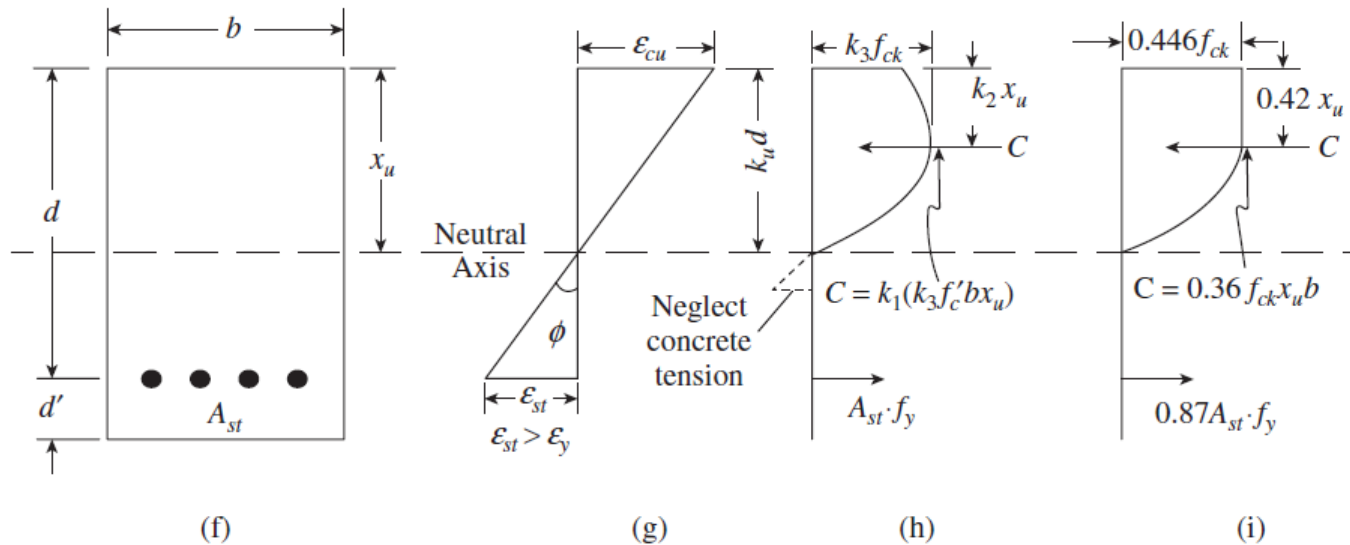


FIG. (f) Cross section of beam (g) Strain distribution at failure (h) Actual stress distribution at failure (i) Assumed stresses as per IS 456

Design of RC beam under flexure

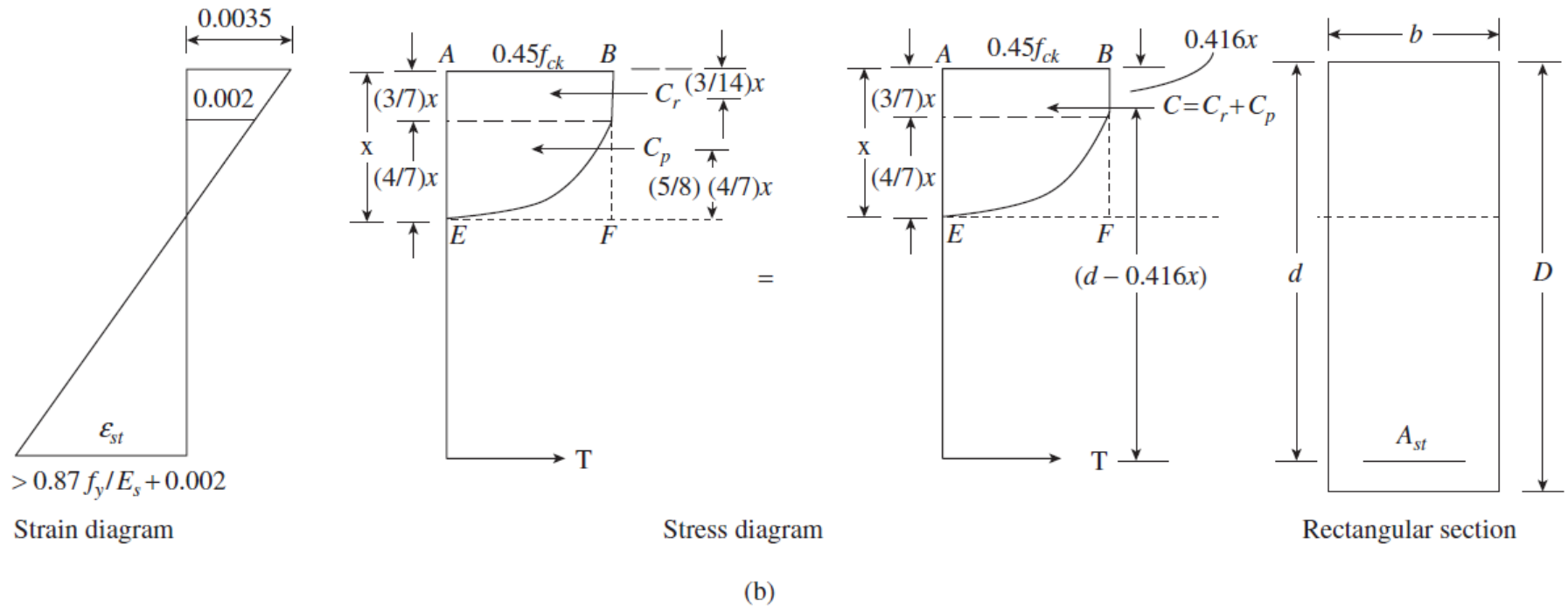
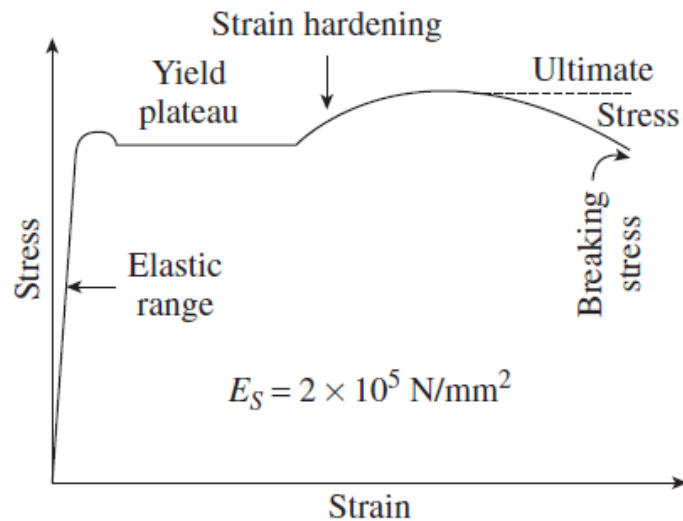
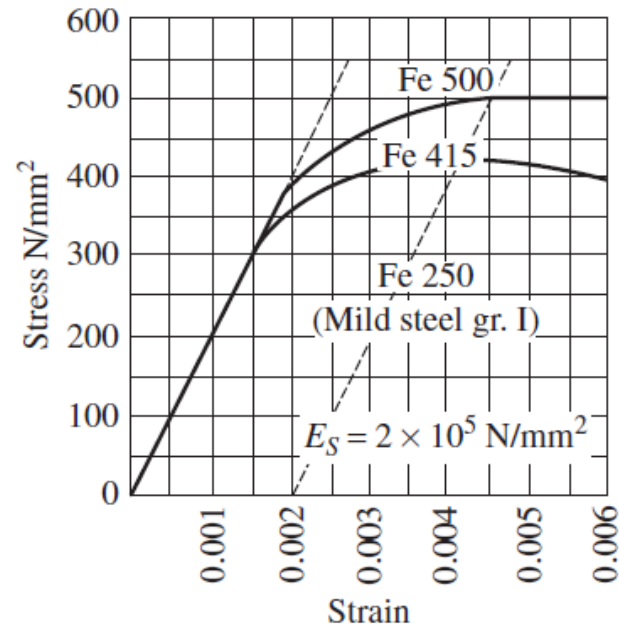


FIG. Stress-Strain curve for concrete (a) Idealized (b) Stress block adopted in IS 456:2000

Stress-Strain Characteristics



(a)



(b)

FIG. Stress-Strain curve (a) Mild steel (b) HYSD bars

Stress–Strain Characteristics

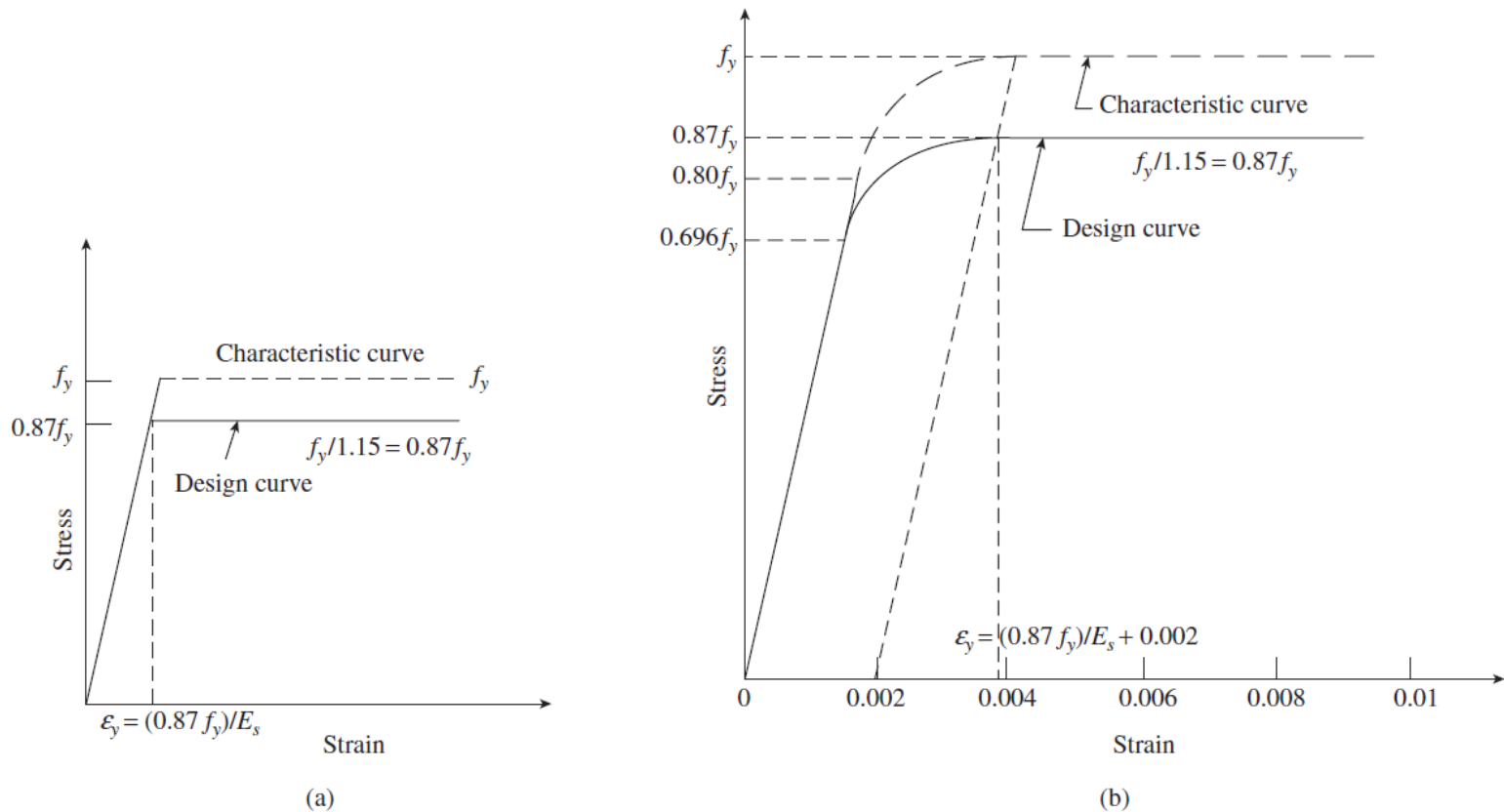


FIG. Stress–Strain curves for steel reinforcements (a) Mild steel bars (b) HYSD bars as per IS 456 ($E_s = 2 \times 10^5 \text{ N/mm}^2$)

Stress-Strain Characteristics

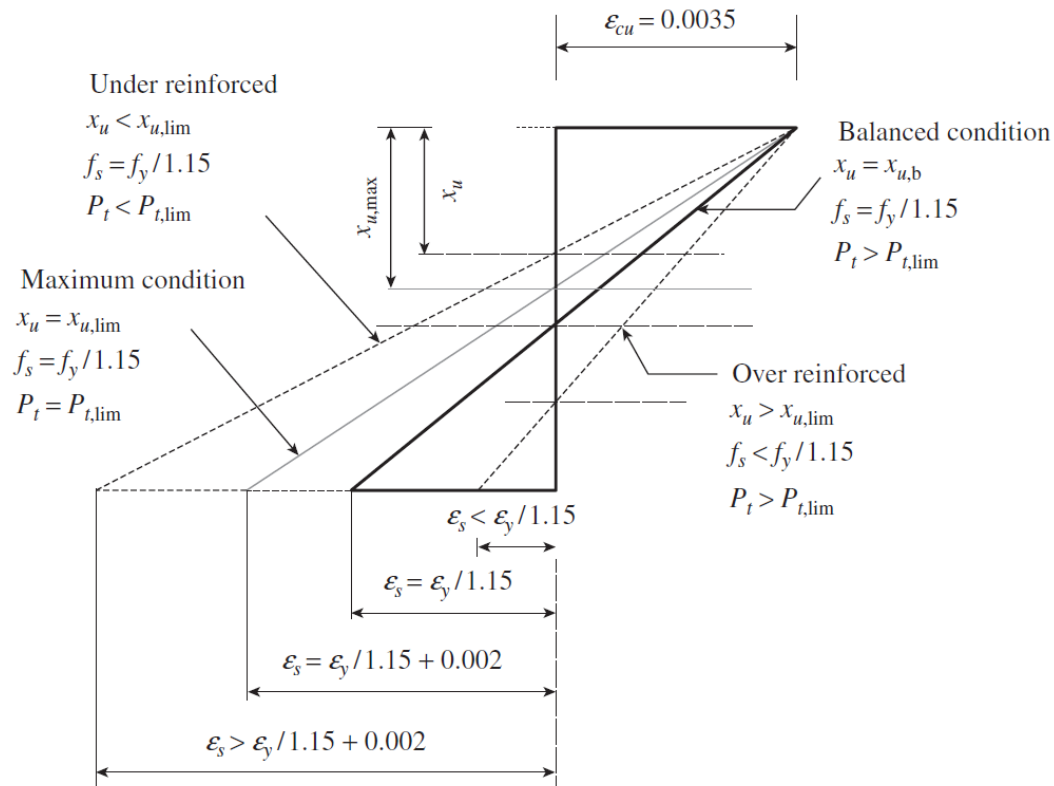
TABLE Design stress at salient strain for HYSD bars

Stress Level	$f_y = 415 \text{ N/mm}^2$		$f_y = 500 \text{ N/mm}^2$	
	Strain $\times 10^{-3}$	Stress (N/mm ²)	Strain $\times 10^{-3}$	Stress (N/mm ²)
0.0	0.00	0.0	0.0	0.0
$0.80f_{yd}$	1.444	288.7	1.74	347.8
$0.85f_{yd}$	1.63	306.7	1.95	369.6
$0.90f_{yd}$	1.92	324.8	2.26	391.3
$0.95f_{yd}$	2.41	342.8	2.77	413.0
$0.975f_{yd}$	2.76	351.8	3.12	423.9
$1.0f_{yd}$	≥ 3.81	360.9	≥ 4.17	434.8

Note: Linear interpolation may be done for intermediate values.

$$f_{yd} = 0.87f_y$$

Neutral axis for balanced, under- and over-reinforced sections



Limiting values of $x_{u,lim}/d$

Type of Steel	f_y (N/mm ²)	$(x_u/d)_{limit}$
Mild steel	250	0.531
HYSD	415	0.479
HYSD	500	0.456

Percentage of limiting steel areas for balanced section

Grade of Steel	(x_u/d) Limit	Limiting Percentage of Steel for Grades M20 to M50
Fe 250	0.531	1.758–4.396
Fe 415	0.479	0.955–2.389
Fe 500	0.456	0.755–1.887

Minimum Tension Reinforcement

- IS 456, f_{ck} might have been assumed to be 25 MPa, and the equation is given in Clause 26.5.1.1 as

$$\frac{A_s}{b_w d} = \frac{0.17\sqrt{f_{ck}}}{f_y} \qquad \frac{A_s}{b_w d} = \frac{0.85}{f_y}$$

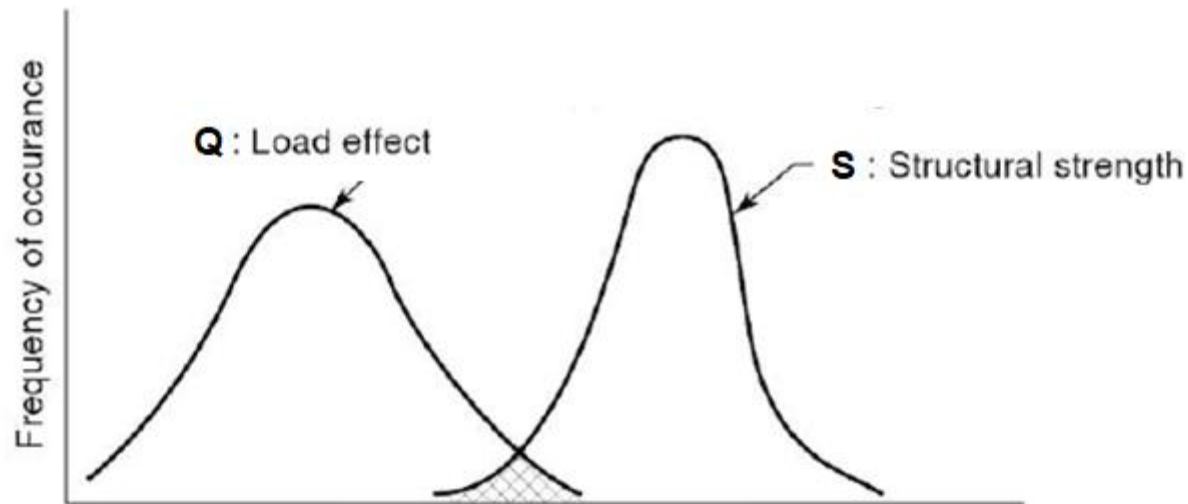
- Clause 6.2.1 of IS 13920:

$$\frac{A_s}{b_w d} = \frac{0.24\sqrt{f_{ck}}}{f_y}$$

Maximum Tension Reinforcement

- IS 456 stipulates that the maximum percentage of tension reinforcement in flexural members be four per cent, which is very high (if both tension and compression steel are provided, it amounts to 8%).
- It should be noted that Clause 6.2.2 of IS 13920 suggests a percentage of steel of 2.5 per cent, which is also high.

Limit state method



Theoretical frequency distribution curves for factored load effects and factored resistance

Design action $\leq$ Design strength

Design Actions

The Design Actions, Q_d , is expressed as $Q_d = \sum_k \gamma_{fk} Q_{ck}$

where

γ_{fk} = partial safety factor for different loads k , to account for:

- a) Possibility of unfavourable deviation of the load from the characteristic value,
- b) Possibility of inaccurate assessment of the load,
- c) Uncertainty in the assessment of effects of the load, and
- d) Uncertainty in the assessment of the limit states being considered.

The loads or load effects shall be multiplied by the relevant γ_f factors, to get the design loads or design load effects.

Design Strength

The Design Strength, S_d , is obtained as given below from ultimate strength, S_u and partial safety factors for materials, γ_m .

$$S_d = S_u / \gamma_m$$

where partial safety factor for materials, γ_m account for:

- a) Possibility of unfavourable deviation of material strength from the characteristic value,
- b) Possibility of unfavourable variation of member sizes,
- c) Possibility of unfavourable reduction in member strength due to fabrication and tolerances, and
- d) Uncertainty in the calculation of strength of the members.

Partial Safety Factors for Loads

Table 18 Values of Partial Safety Factor γ_f for Loads
(Clauses 18.2.3.1, 36.4.1 and B-4.3)

Load Combination	Limit State of Collapse			Limit States of Serviceability		
	<i>DL</i>	<i>IL</i>	<i>WL</i>	<i>DL</i>	<i>IL</i>	<i>WL</i>
(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>DL + IL</i>	1.5		1.0	1.0	1.0	–
<i>DL + WL</i>	1.5 or 0.9 ¹⁾	–	1.5	1.0	–	1.0
<i>DL + IL + WL</i>	1.2			1.0	0.8	0.8

NOTES

- 1 While considering earthquake effects, substitute *EL* for *WL*.
 - 2 For the limit states of serviceability, the values of γ_f given in this table are applicable for short term effects. While assessing the long term effects due to creep the dead load and that part of the live load likely to be permanent may only be considered.
- ¹⁾ This value is to be considered when stability against overturning or stress reversal is critical.

Limit state of serviceability

- Deflection
- Crack
- Vibration